viva-genesis

Full Modeling Plan

A Process Bigraph architecture for modeling the transition from geochemistry to cells

Central design principle: Do not start with a cell. Build a geological world in which increasingly biological forms of organization must emerge. The Process Bigraph should itself change as localization, boundaries, growth, heredity, and ecology appear.

Modeling progression

Geochemistry -> electrochemistry -> reaction networks -> reaction-transport -> compartments -> bioenergetics -> heredity -> ecology -> autonomy

1. Architecture at a glance

viva-genesis should be a composition of model classes whose organization changes as the simulated world becomes more biological. The same environmental world persists throughout; new processes and bounded agents are added by structural transitions rather than replacing the earlier model.

Stage	Transition	Primary model types	Biological meaning
0	Rock -> vent	Geochemical speciation/equilibrium; serpentinization kinetics; reactive transport; mineral precipitation/dissolution; porosity/permeability	A persistent source of chemical disequilibrium
I	Vent -> reduced carbon	Electrochemistry; mineral-surface catalysis; microkinetics; redox thermodynamics	Geological energy is captured in organic chemistry
II	Reduced carbon -> protometabolism	Kinetic reaction networks (ODE/SBML); thermodynamic networks; autocatalytic networks; optional constraint-based models	Persistent connected chemical organization
III	Distributed -> localized chemistry	Advection-diffusion-reaction PDEs; thermophoresis; adsorption; pore-scale transport	Local chemical individuality
IV	Localization -> protocells	Amphiphile assembly; permeability; osmosis; compartment mechanics; growth/division	A bounded individual
V	Environmental -> internal energy	Membrane electrochemistry; ion transport; redox coupling; energy-budget models	Partial energetic autonomy
VI	Persistence -> heredity	Polymerization/replication kinetics; stochastic inheritance; mutation/error; compositional inheritance; evolutionary models	Ancestry begins predicting phenotype
VII	Individuals -> ecology	Spatial agent-based; resource competition; cross-feeding; ecological networks; horizontal exchange	Selection among interacting lineages
VIII	Vent-dependent -> autonomous	Integrated protocell model; perturbation; ablation; viability/robustness	Functions supplied by geology become internalized

Cross-cutting layers: C-N-P chemistry, spatial environmental fields, free-energy accounting, parameter provenance, and the dynamic Process Bigraph structure should persist across stages.

2. Process Bigraph design

Process Bigraphs should represent biology, not merely software wiring. Early in the simulation there is no cell object: only environmental stores and processes. Localization creates persistent chemical regions; boundary

formation creates a protocell composite; division rewrites one individual into two; heredity adds inherited state; ecology embeds many individuals in shared fields.

Layer	Canonical state	Processes
Physical world	Geometry; minerals; temperature; pressure; pH; ionic strength; H2/CO2; velocity; redox/electric fields	Serpentinization, transport, precipitation/dissolution, electrochemistry
Chemical organization	Molecular pools; catalysts; reaction fluxes; C/N/P resources; primitive energy currencies	Carbon fixation, nitrogen chemistry, phosphorus chemistry, autocatalytic networks, amphiphile synthesis
Biological individuals	Boundary; internal chemistry; energy state; volume; growth/division state; inherited state	Membrane exchange, metabolism, energy capture, growth, division, heredity
Ecology	Lineage IDs; spatial positions; uptake/secretion; interaction fluxes; population state	Competition, cross-feeding, parasitism, cooperation, horizontal exchange

Structural rewrites

- Boundary formation: amphiphile concentration and local persistence cross assembly criteria -> create a protocell composite around localized chemistry.
- Growth: internal production and osmotic balance increase volume and membrane area.
- Division: a physical instability or size criterion rewrites one protocell into two daughters with stochastic partitioning of molecules, catalysts, membrane components, and hereditary material.
- Extinction: sustained negative energy/maintenance balance dissolves the protocell back into environmental material.
- Heredity transition: an inheritance process is attached only when precursor availability, polymerization/copying chemistry, and fidelity support it.

3. Stage 0 - Rock to vent

Objective: create a physically credible, evolving hydrothermal reactor before making any origin-of-life claim.

Model class	Role	Suggested implementation pattern
Geochemical speciation/equilibrium	Fluid composition, pH, redox/speciation, saturation states	PHREEQC as the geochemical core
Serpentinization / water-rock kinetics	H2 generation, alkalinity, Fe redox, reaction progress	Kinetic process coupled to mineral inventory
Reactive transport	Advection/diffusion of dissolved species through pores	FiPy/spatio-flux-style spatial fields
Mineral precipitation/dissolution	FeS/carbonate/mineral phase changes	Geochemistry process updates mineral stores
Porosity/permeability	Feedback of mineral change on flow and residence time	Dynamic geometry/permeability process

Key test: Can H2 production, high pH, catalytic mineral phases, and open transport persist together, or does the reactor poison or seal itself?

4. Stage I - Vent to reduced carbon

Objective: turn geological disequilibrium into measurable organic synthesis without hiding the mechanism in a generic gradient-efficiency parameter.

Model class	State/outputs	Questions
Electrochemical model	Electrode/redox potentials, conductivity, current density	Can the mineral interface actually deliver electrons at a useful rate?
Mineral-surface catalysis	Active sites, mineral identity, Ni/Fe/S composition, passivation	Which surfaces enable or suppress CO2 reduction?
Microkinetic surface model	Adsorption, elementary steps, rate constants, product flux	Which kinetic bottleneck controls formate/acetate/pyruvate production?
Thermodynamic model	Delta G and reaction quotients under local conditions	Is failure thermodynamic or kinetic/electrical?

PBG composition: fluid chemistry -> redox potential -> mineral conductivity/current -> electron transfer -> surface kinetics -> reduced-carbon pools.

5. Stage II - Reduced carbon to protometabolism

Objective: allow connected, persistent reaction organization to emerge from a broader chemical landscape rather than inserting a modern metabolic pathway as the answer.

Model class	Use	Candidate libraries / representation

Kinetic reaction network	C1/C2/C3 chemistry and continuous-flow dynamics	SBML + Tellurium/COPASI/libRoadRunner via reusable PBG wrappers
Thermodynamic network	Filter/weight reactions by local energetic feasibility	eQuilibrator-like thermodynamic calculations
Catalyst availability model	Make rates conditional on mineral/metal/catalyst state	Shared catalyst store linked to Stage I
Autocatalytic/network analysis	Detect self-reinforcing motifs, persistence, connectivity	Graph/network analysis over realized fluxes
Constraint-based model, optional	Explore larger feasible chemical spaces when kinetics are unavailable	FBA-like formulation used cautiously for prebiotic chemistry

Core experiment: Expose a broad network to the same continuous-flow environment and ask which subnetworks dominate, persist, and amplify. Compare emergent motifs with Wood-Ljungdahl-like and rTCA-like organization only after the fact.

6. Cross-cutting chemistry - C, N, P and proto-energetics

Module	Model type	Key outputs
Carbon	Kinetic + thermodynamic reaction network	Reduced C flux, carbon-fate distribution, network persistence
Nitrogen	N2/NO3-/NO2- reduction and assimilation kinetics	NH3/NH4+ supply, amino-acid precursor flux, C:N limitation
Phosphorus	Speciation, adsorption/precipitation, activation kinetics	Available/reactive phosphate, loss to minerals, C:N:P overlap
Primitive energy currencies	Coupled-reaction / energy bookkeeping model	Thioester/acyl-phosphate/poly-P/redox pools; stored usable work
Resource limitation	Shared resource-budget model	Which resource first limits persistence/growth?

New scientific gate: Can the same local environment support coupled C-N-P chemistry and capture environmental free energy into transferable chemical work?

7. Stage III - Localization and chemical individuality

Objective: determine whether chemistry can persist locally despite diffusion, washout, degradation, and heterogeneous transport.

Model class	Mechanism	PBG role
Advection-diffusion-reaction PDE	Flow, molecular diffusion, local reaction	Shared spatial field world
Thermophoresis	Species-specific motion in temperature gradients	Transport term coupled to temperature field
Adsorption/desorption	Mineral-surface retention and release	Surface stores coupled to dissolved pools
Degradation/leakage	Chemical destruction and loss	Competes directly with retention
Dynamic pore geometry	Irregular confinement and evolving residence times	Inherited from Stage 0 mineral/geometry state

Biological criterion: Individuality begins when local chemical states have persistent histories: retention and regeneration outlast destruction and exchange with the surrounding world.

8. Stage IV - Protocells, membranes, growth and division

Objective: make the transition from localized chemistry to a bounded, physically reproducing individual.

Model class	Mechanism	State
Amphiphile synthesis network	Endogenous fatty-acid/alkanol/thioester/lipid precursor production	Membrane precursor pools
Assembly model	Critical aggregation and mixed-amphiphile vesicle formation	Boundary creation event
Permeability model	Species-specific exchange across boundary	Permeability coefficients and fluxes
Osmotic model	Solute imbalance drives water influx/efflux	Volume, pressure, lysis risk
Compartment mechanics	Membrane area, shape/size constraints, instability	Growth and division criteria
Partition model	Stochastic daughter allocation	Internal molecules, catalysts, membrane, inherited state

PBG rewrite: A localized chemical region becomes a protocell composite only when assembly criteria are satisfied. Division creates daughter composites; lysis returns internal material to the environment.

9. Stage V - Internal bioenergetics

Objective: model the gradual internalization of energetic functions initially supplied by geology.

Model class	Represents	Measurements

Membrane electrochemistry	Delta pH, membrane potential, ion gradients	Proton/Na+ motive force
Ion transport	Passive leakage and primitive transport/pumping	Ion flux and energetic cost
Redox coupling	Electron donors/acceptors and primitive carriers	Redox balance and coupled flux
Chemical energy model	Thioesters, acyl phosphate, poly-P or analogous currencies	Stored chemical work
Energy budget	Capture, maintenance, synthesis, transport, growth, dissipation	Fraction of environmental free energy converted to usable work

Autonomy metric: Track the fraction of essential energetic work supplied internally rather than directly by the external vent gradient.

10. Stage VI - Heredity and evolving information

Objective: ask when high-fidelity templated heredity becomes advantageous over cheaper forms of chemical memory.

Inheritance model	Model type	What is inherited
Mineral-state	Structural/state inheritance	Catalytic surface or mineral configuration
Compositional	Stochastic partition + attractor dynamics	Molecular/catalyst composition
Network regeneration	Autocatalytic reaction-network dynamics	Self-maintaining chemical organization
Peptide-like	Polymerization/replication kinetics	Catalytic polymer composition
RNA-like	Template replication + mutation/error model	Sequence information
Hybrid	Coupled metabolic + templated inheritance	Chemical organization plus digital information

Key test: Under common resource and energy budgets, when does the benefit of fidelity and information capacity exceed the cost of replication? Measure when ancestry becomes predictive of phenotype.

11. Stage VII - Ecology before LUCA

Objective: embed many protocells in the same spatial reaction-transport world and let ecological interactions arise through real resource exchange.

Model class	Mechanism	Emergent question
Spatial agent-based model	Movement, growth, division, death in pore network	Can lineages coexist spatially?
Shared resource fields	Uptake/secretion coupled to PDE environment	Does cross-feeding stabilize otherwise impossible chemistry?
Competition model	Resource and space limitation	Which organizations dominate?
Parasitism/cooperation	Asymmetric exchange and public goods	When are collective functions stable?
Horizontal exchange	Transfer of catalysts/polymers/inherited material	Can innovations spread before modern genetics?
Ecological network analysis	Interaction and flux graphs	Does community-level function precede cell-level completeness?

LUCA use: Treat reconstructed LUCA capabilities as a distant constraint or comparison, not as the target architecture of the first replicator.

12. Stage VIII - Escape and autonomy

Objective: determine which functions must be internalized before an evolving lineage can persist when geological support is withdrawn.

Experiment	Perturbation	Readout
Gradient ablation	Flatten external pH/electrochemical gradient	Persistence, energy balance, reproduction
H2 ablation	Reduce/remove geochemical electron donor	Alternative energy dependence
Mineral ablation	Remove catalytic/conductive interfaces	Catalytic independence
Confinement ablation	Open pores / increase dilution	Boundary and transport autonomy
Feed ablation	Reduce externally supplied precursors	Biosynthetic completeness
Innovation ablation	Remove evolved internal mechanisms one by one	Minimum sufficient innovation set

Final metric: Autonomy is continuous: quantify how many essential functions remain environmentally supplied versus internally maintained. A cell is earned by surviving progressive removal of the geological scaffold.

13. Reusable Process Bigraph composites

Composite	Contains
geochemical_world	Serpentinization + speciation + mineral evolution + reactive transport
electrochemical_interface	Redox potentials + conductivity/current + mineral catalysis
prebiotic_chemistry	Carbon + nitrogen + phosphorus + primitive energy currencies
pore_localization	Advection/diffusion + thermophoresis + adsorption + degradation
proto_boundary	Amphiphile synthesis + assembly + permeability + osmosis
proto_individual	Prebiotic chemistry + boundary + growth + division
proto_bioenergetics	Membrane electrochemistry + ion transport + chemical energy coupling
proto_heredity	Interchangeable compositional / network / peptide / RNA-like inheritance
proto_ecology	Many proto-individuals + shared spatial fields + uptake/secretion + interactions

14. Existing libraries and infrastructure to exploit

Need	Existing ecosystem / library direction	Use in viva-genesis
Geochemical chemistry	PHREEQC / phreeqpython	Speciation, mineral saturation, geochemical reactions
Spatial transport	FiPy; spatio-flux patterns	Shared reaction-transport world and agent-field coupling
Kinetic networks	SBML; libRoadRunner; Tellurium; COPASI; viva-biomodels patterns	Reusable chemical-network processes
Thermodynamics	eQuilibrator-style calculations	Reaction feasibility and free-energy bookkeeping
Agent/composite patterns	process-bigraph; Vivarium whole-cell / multiscale patterns	Hierarchical protocell composites and structural rewrites
Physical agent mechanics	viva-munk-style mechanics where appropriate	Compartment movement/collision or later cell-scale mechanics
Visualization	bigraph-viz + Workbench visualization contracts	Show both world state and evolving causal architecture
Campaign management	Vivarium Workbench investigations/studies	Parameter sweeps, gates, provenance, reproducibility

15. Canonical biological schemas

Simulator-native variables should be adapted into shared biological concepts. This keeps the composition intelligible and makes models substitutable.

Schema	Representative state
environment	temperature, pressure, pH, ionic strength, flow
minerals	phase abundance, surface area, catalytic sites, conductivity, porosity
molecules	CO2, H2, formate, acetate, pyruvate, NH4+, phosphate, amphiphiles, polymers, etc.
energy	redox potential, electric potential, ion gradients, stored chemical work, dissipated energy
boundary	area, composition, permeability, charge, integrity
individual	volume, internal/external identity, maintenance flux, growth rate, division state, persistence
heredity	inherited state, sequence/composition, fidelity, mutation rate, information content
lineage/ecology	parent ID, lineage ID, spatial position, uptake/secretion, interaction state

16. Model substitution as a scientific experiment

A major advantage of the Process Bigraph architecture is that uncertainty in mechanism can be represented by swapping models while holding the surrounding world constant.

Interface	Alternative models to compare
carbon_fixation	Empirical kinetics; explicit mineral-surface microkinetics; electrochemical current-limited model; alternative catalysts
proto_metabolism	Curated kinetic network; larger thermodynamic/kinetic network; autocatalytic-set model; constraint-based approximation
localization	Idealized thermophoretic trap; irregular pore PDE; adsorption-dominated retention; mixed mechanisms
boundary	Single-chain amphiphiles; mixed amphiphiles; thioester/aminothiol-derived membranes
heredity	Compositional; network; peptide-like; RNA-like; hybrid
environment	Deep alkaline vent; shallow vent; terrestrial geothermal; generic nonequilibrium control

Scientific benefit: The campaign can distinguish uncertainty in parameters from uncertainty in model structure. Competing origin scenarios can be tested through the same interfaces and behavioral gates.

17. Experimental design and acceptance

- Use phase maps rather than one successful trajectory: every transition should have pass and failure regions.
- Carry parameter provenance explicitly: measured, experimentally calibrated, inferred, extrapolated, or hypothetical.
- Maintain matter, charge, and free-energy accounting across model boundaries.
- Use mechanism ablations to establish necessity, not just correlation.
- Use adversarial controls: generic nonequilibrium chemistry should not automatically pass vent-specific gates.
- Make negative results first-class outputs. A broken transition identifies a concrete experimental bottleneck.
- Keep world view, mechanism view, and decision view synchronized in the Workbench for every study.

18. Recommended implementation sequence

Priority	Build	Why first
1	Canonical schemas + geochemical_world	Establishes common state and credible physical world
2	electrochemical_interface	Replaces abstract gradient efficiency with explicit causal mechanism
3	Expanded prebiotic_chemistry (C-N-P + proto-energetics)	Makes resource limitation and energy coupling testable
4	pore_localization on shared spatial fields	Tests whether chemistry can become locally persistent
5	proto_boundary + endogenous amphiphile production	Creates the first explicit bounded individual
6	growth/division structural rewrites	Turns chemistry into differential physical reproduction
7	competing heredity modules	Tests when information becomes worth its cost
8	proto_ecology	Lets collective function emerge before complete cells
9	autonomy/ablation campaign	Determines what must be internalized to leave the vent

Near-term milestone: Can a physically plausible hydrothermal pore transform inorganic geological feedstocks into a growing, dividing chemical individual without presupposing cellular machinery?

19. The conceptual endpoint

The strongest version of viva-genesis does not define a cell in advance. It represents the world before cells and asks when a persistent boundary begins to contain processes that were previously distributed across geology and environment. In this architecture, the emergence of a cell is both a biological transition and a change in the Process Bigraph itself.

Cells arise when processes previously distributed across the environment become composed inside a persistent, reproducing, heritable boundary.